Meeting Module

Names:	Gabriele, Luis
Topic(s):	Electronics
Date:	5/11/26

Was there a final file made Y/N: N

Where was it uploaded (GITHUB/DRIVE):

What did you do in the meeting?

Gabriele and Luis had a meeting in which they talked about some of the safety structure of the electronic circuit. They decided to create an electronic switch to toggle deliberately via the ESP32 PINS, this can help the user access the vacuum at will. We had two options between the High side switch and the Low Side switch and after testing both we preferred to continue experimenting on the high side switch. This was because it was a more flexible approach to our circuit. Not only it was easier to add onto the PCB but it was even more intuitive to organize the circuit better. However, the high side took more time and components to include in the electronics which implies a greater cost. The high side switch consists of two transistors, a more potent resistant p-channel mosfet that drives the entire circuit (name is in the E-Lab) and a smaller n-channel mosfet that drives the bigger mosfet down letting current pass through it. It seems very neat and experimenting with a known load, the currents measured seemed reliably close to our expectations. I.e. with a 100ohms resistor at 10 volts, we were close to a current of 0.1 amps which is to be expected.

The thermistor chosen is present in the E-lab and it can be used as a simple ADC working thermistor at 5 volts, more research has to be done.

Next meeting, there should be research about the fuse and the killswitch to integrate in the circuit.

Furthermore, Some shunt resistors have been found. They have been experimentally measured and the sum approximates to 12milliohms which is close to our goal of 9 milliohms.